could easily accumulate more than their ten percent share of helium, but, as predicted, no one has ever found a region of the galaxy with less.

Some thirty years before it was discovered and isolated on Earth, astronomers detected helium in the spectrum of the Sun's corona during the total eclipse of 1868. As noted earlier, the name helium was duly derived from Helios, the Greek sun god. And with 92 percent of hydrogen's buoyancy in air, but without its explosive characteristics, helium is the gas of choice for the outsized balloon characters of the Macy's Thanksgiving Day parade, making the department store second only to the U.S. military as the nation's top user of the element.

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Lithium is the third simplest element in the universe, with three protons in its nucleus. Like hydrogen and helium, lithium was made in the big bang, but unlike helium, which can be manufactured in stellar cores, lithium is destroyed by every known nuclear reaction. Another prediction of big bang cosmology is that we can expect no more than one percent of the atoms in any region of the universe to be lithium. No one has yet found a galaxy with more lithium than this upper limit supplied by the big bang. The combination of helium's upper limit and lithium's lower limit gives a potent dual-constraint on tests for big bang cosmology.

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The element carbon can be found in more kinds of molecules than the sum of all other kinds of molecules combined. Given the abundance of carbon in the cosmos—forged in the cores of stars, churned up to their surfaces, and released copiously into the galaxy—a better element does not exist on which to base the chemistry and diversity of life. Just edging out carbon in abundance rank, oxygen is common, too, forged and released in the remains of exploded stars. Both oxygen and carbon are major ingredients of life as we know it.

But what about life as we don't know it? How about life based on the element silicon? Silicon sits directly below carbon on the Periodic Table, which means, in principle, it can create the same portfolio of molecules that carbon does. In the end, we expect carbon to win because it's ten times more abundant than silicon in the cosmos. But that doesn't stop